Detailed Theory: Router IOS & Security Device Manager (SDM)

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1. What is Router IOS?

- **IOS (Internetwork Operating System)** is Cisco’s proprietary operating system used on most Cisco routers and switches.
- It provides a command-line interface (CLI) for configuring and managing Cisco devices.
- IOS is a **monolithic** OS, highly modular, optimized for network routing and switching.

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2. Functions of IOS

- Routing protocols (e.g., OSPF, EIGRP, BGP)
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- Packet forwarding and switching
- Access control and firewall features
- Network Address Translation (NAT)
- VPN services and security
- Interface and device management

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3. IOS Command Modes

- **User EXEC Mode:** Basic commands, limited access (prompt: Router>)
- **Privileged EXEC Mode:** More commands, can view configs (prompt: Router#)

- **Global Configuration Mode:** Configuration commands apply globally (prompt: Router(config)#)
- **Interface Configuration Mode:** Config commands for specific interfaces (prompt: Router(config-if)#)
- Others: line config, router config, etc.

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4. IOS Security Features

- Access Control Lists (ACLs) to filter traffic
- Authentication, Authorization, and Accounting (AAA)
- Secure management (SSH, HTTPS)
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VPN support (IPSec)

- Logging and monitoring tools

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5. What is Cisco Security Device Manager (SDM)?

- SDM is a **graphical user interface (GUI) tool** for Cisco routers.
- It simplifies router configuration by providing a web-based or desktop interface.
- Especially useful for administrators unfamiliar with CLI.
- SDM allows configuration of security policies, VPNs, firewall rules, NAT, and QoS.

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6. SDM Features

- Step-by-step wizards for common tasks
- Visual representation of network and policies
- Security audit tools
- Easy access to monitoring and troubleshooting tools
- Supports router IOS features through an intuitive UI

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7. How SDM Works

- Runs on a PC and connects to the router via HTTP or HTTPS.

- The router must have HTTP server enabled to support SDM access.
- SDM converts GUI inputs into IOS commands and applies them to the router.

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8. Benefits of Using SDM

- Simplifies complex router configurations
- Reduces chances of misconfiguration
- Speeds up deployment of network security policies
- Provides visual feedback and audit reports

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9. Limitations of SDM

- Limited support on newer routers and IOS versions (Cisco is moving towards Cisco DNA Center and CLI)
- Some advanced configurations may still require CLI
- Performance depends on PC and network connection quality

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10. Basic IOS Security Commands Examples

- Enable password:
  Router(config)# enable secret [password]
- Configure SSH:

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```
Router(config)# ip domain-name example.com   Router(config)# crypto key generate rsa   Router(config)# line vty 0 4 Router(config-line)# transport input ssh   Router(config-line)# login local
```

- Configure ACL:

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```
Router(config)# access-list 100 permit tcp any any eq 80   Router(config)# interface GigabitEthernet0/0   Router(config-if)# ip access-group 100 in
```